

Md Arafat Hossen Rabby

COMPUTER SCIENCE GRADUATE | AI / MACHINE LEARNING | DATA SCIENCE

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[LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in **Python, SQL, machine learning, deep learning, data analysis, and computer vision**. Developed and evaluated machine learning systems using **PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, and Transformers**. BSc thesis research focused on **few-shot learning for acne severity classification**, including lesion-aware feature representation, FiLM-based feature calibration, and Per-Channel Variance Weighting (PCVW). Interested in **AI/ML, Data Science, Computer Vision, and applied machine learning systems**.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Analysis, Data Visualization

Machine Learning: Scikit-learn, Machine Learning, Deep Learning, Transfer Learning

AI & Computer Vision: PyTorch, TensorFlow, Computer Vision, Few-Shot Learning, EfficientNet, FiLM

Natural Language Processing: NLP, Transformers, DistilBERT, TF-IDF, Text Classification

Tools & Deployment: Git, GitHub, Streamlit, Flask, VS Code, Ubuntu

Databases: MySQL, MongoDB

EDUCATION

Premier University, Chittagong

Bachelor of Science in Computer Science and Engineering

• **CGPA: 3.21 / 4.00**

Chittagong, Bangladesh

Nov 2021 – Aug 2026

Omargoni M.E.S College

Higher Secondary Certificate (HSC), Science

• **GPA: 4.17 / 5.00**

Chittagong, Bangladesh

Jul 2019 – May 2020

RESEARCH / THESIS

DLR-PCVW — Few-Shot Acne Severity Classification

[GitHub](#)

Python, PyTorch, EfficientNet-B0, Computer Vision, Few-Shot Learning, FiLM, PCVW

- Developed a few-shot deep learning framework for **4-class acne severity classification** under limited labelled data.
- Implemented lesion-aware feature representation, **FiLM-based feature calibration**, and **Per-Channel Variance Weighting (PCVW)** for prototype-based classification.
- Evaluated model performance under **1-, 3-, 5-, and 10-shot** learning settings.
- Achieved **80.27% ± 0.43% accuracy** in the 10-shot setting across multiple evaluation runs.
- Built an interactive **Streamlit** demonstration for model inference and visualization.

PROJECTS

Email Spam Detection — DistilBERT & Machine Learning

[GitHub](#)

TF-IDF, DistilBERT, Keras, PyTorch, Scikit-learn, Pandas, NumPy

- Developed an NLP classification system to classify emails as **Spam or Ham** using traditional machine learning and Transformer-based methods.
- Implemented and compared **TF-IDF with traditional ML models** against a pretrained **DistilBERT** classifier.
- Used a pretrained Transformer model from **Hugging Face** for text classification.
- Built an interactive **Streamlit** application providing predictions and confidence scores.

Car Price Prediction

[GitHub](#)

Python, Pandas, NumPy, Scikit-learn, Random Forest, Machine Learning

- Developed a supervised machine learning regression model to predict used-car prices from vehicle attributes.
- Performed data preprocessing, exploratory analysis, feature preparation, model training, and evaluation using Python.
- Implemented a **Random Forest Regressor** and evaluated predictive performance using regression metrics.

Diabetes Prediction Web Application

[GitHub](#)

Python, Scikit-learn, Flask, Pandas, Machine Learning

- Developed a machine learning-based web application for diabetes risk prediction using structured health-related features.
- Implemented data preprocessing and machine learning model training using **Python, Pandas, and Scikit-learn**.
- Integrated the trained model into a **Flask** web application for interactive predictions.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Completed hands-on learning in **Python, Machine Learning, Data Science, NLP, and Computer Vision** through academic and independent projects.
- Actively developing skills in **SQL, Data Analysis, Data Visualization, and Power BI** for graduate-level Data Science and Data Analyst roles.